



Introduction to Artificial Intelligence (INTRAI)

Module 3: Machine Learning Basics

Training, Overfitting and Bias

Lesson Notes

KMVTECHUG

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Training and Testing

We train on one set of data and test on another set the model has not seen. If it only memorised training examples, test performance will drop.

That is why serious teams hold out test data instead of reporting scores only on the training set.

Overfitting in Simple Terms

Overfitting means the model fits training data too tightly, including noise, and fails on new cases.

It is like a student who memorises past paper answers word-for-word but cannot handle a rephrased question.

Data Quality and Bias

If one group is under-represented in training data, the model may perform worse for that group. Bias is not always intentional, but impact is real.

In local settings, language, lighting, device quality, and who appears in the data all matter.

Reflect: *How is overfitting similar to a study habit you want to avoid before exams?*

Closing

Module 3 covered ML types, train/test splits, overfitting, and bias. Module 4 turns to generative AI and study use.